EVEI LIGHT METER – Arduino Mega 2560 carrier shield

microSD: on-board socket, 3.3V LDO, 74LVC125 shifts MOSI/SCK/CS 5V–>3.3V; MISO (3.3V) goes straight to D50 (same as Adafruit 254 breakout).

Pin map from the team sketch (Lightmeter_Week8.ino): SD CS=D53 (HW SPI 50/51/52), GPS=Serial1 (D18/D19), IMU=Serial2 (D16/D17), I2C=D20/D21,
LEDs common–cathode to GND, active HIGH (setLED() in Week 5–8 sketches: YELLOW = both HIGH): GPS R=D4 G=D7, LUX R=D5 G=D6, IMU R=D8 G=D9; buttons to GND with INPUT_PULLUP: sleep D10, log toggle D11, force log D12.

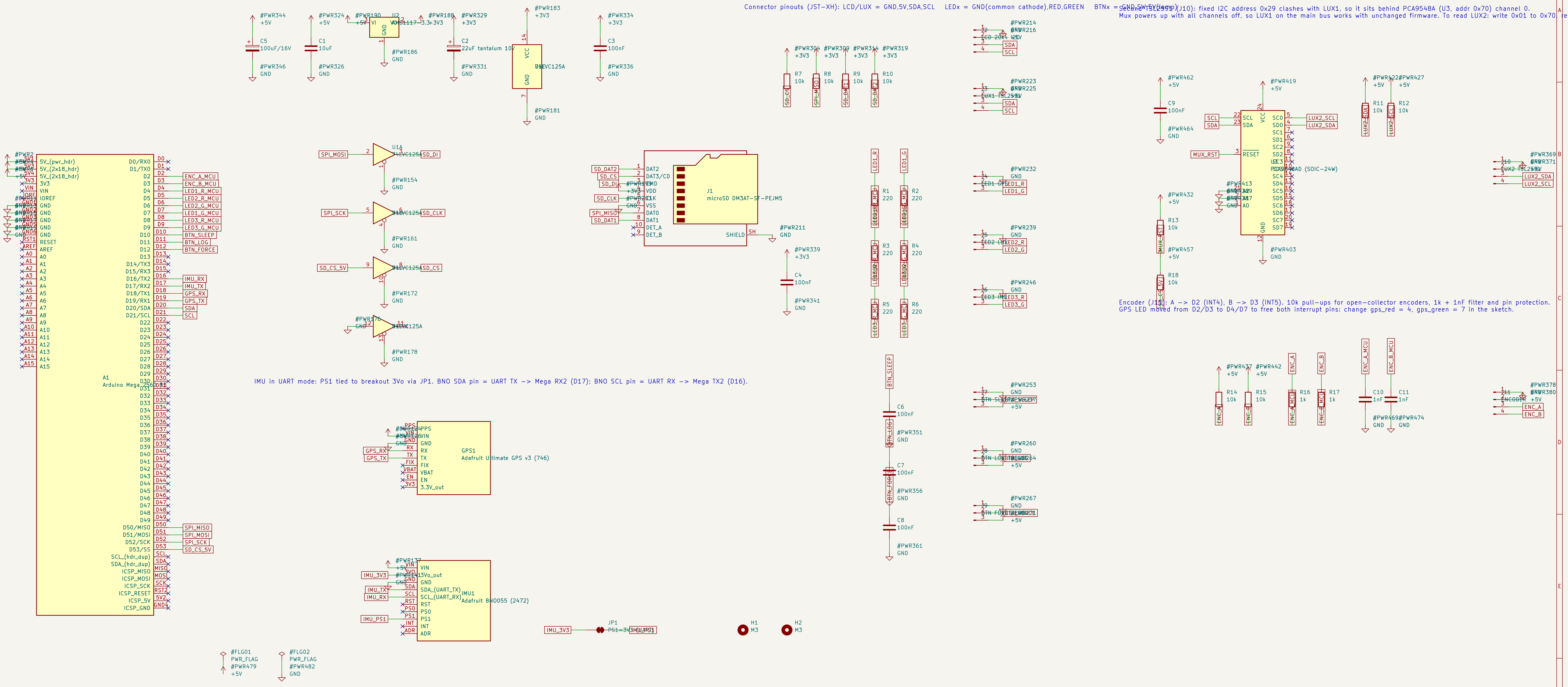
Connector pinouts (JST–XH): LCD/LUX = GND,5V,SDA,SCL   LEDx = GND(common cathode),RED,GREEN   BTNx = GND,SW,5V(lamp)

REV B ADDITIONS

Second TSL2591 (J10): fixed I2C address 0x29 clashes with LUX1, so it sits behind PCA9548A (U3, addr 0x70) channel 0.
Mux powers up with all channels off, so LUX1 on the main bus works with unchanged firmware. To read LUX2: write 0x01 to 0x70, read, wri

Encoder (J19): A –> D2 (INT4), B –> D3 (INT5). 10k pull–ups for open–collector encoders, 1k + 1nF filter and pin protection.
GPS LED moved from D2/D3 to D4/D7 to free both interrupt pins: change gps_red = 4, gps_green = 7 in the sketch.

IMU in UART mode: PS1 tied to breakout 3Vo via JP1. BNO SDA pin = UART TX –> Mega RX2 (D17); BNO SCL pin = UART RX –> Mega TX2 (D16).



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